

# Atomic structure

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## Atomic Structure Study Notes

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### 1. Overview/Introduction

The atomic structure is a fundamental concept in chemistry that describes the composition and arrangement of atoms. Atoms are the building blocks of matter, and understanding their structure is crucial for understanding chemical properties and reactions. In this section, we will explore the key concepts, definitions, and examples related to atomic structure.

### 2. Key Concepts

- **Atomic Number**: The number of protons in an atom's nucleus, which determines the element of an atom.
- **Mass Number**: The total number of protons and neutrons in an atom's nucleus.
- **Isotopes**: Atoms of the same element with different numbers of neutrons.
- **Electron Configuration**: The arrangement of electrons in an atom's energy levels or orbitals.
- **Energy Levels**: The regions around the nucleus where electrons are found.
- **Orbitals**: The specific regions within energy levels where electrons are found.

### 3. Important Definitions

- **Proton**: A positively charged particle found in the nucleus of an atom.
- **Neutron**: A particle with no charge found in the nucleus of an atom.
- **Electron**: A negatively charged particle that orbits the nucleus of an atom.
- **Nucleus**: The central part of an atom that contains the protons and neutrons.
- **Atomic Radius**: The distance from the nucleus to the outermost energy level of an atom.

### 4. Examples and Worked Solutions

#### Example 1: Calculating Atomic Number and Mass Number

| Element | Protons | Neutrons | Atomic Number | Mass Number |
|---------|---------|----------|---------------|-------------|
| Carbon  | 6       | 6        | 6             | 12          |
| Oxygen  | 8       | 8        | 8             | 16          |

#### Worked Solution:

- The atomic number of carbon is 6, which means it has 6 protons in its nucleus.
- The mass number of carbon is 12, which means it has 6 protons and 6 neutrons in its nucleus.

#### Example 2: Electron Configuration

- The electron configuration of sodium (Na) is  $1s^2 2s^2 2p^6 3s^1$ .
- The electron configuration of chlorine (Cl) is  $1s^2 2s^2 2p^6 3s^2 3p^5$ .

### 5. Key Points to Remember

- The atomic number determines the element of an atom.

- Isotopes have the same atomic number but different mass numbers.
- Electron configuration determines the chemical properties of an element.
- Energy levels and orbitals are important in understanding electron configuration.

## 6. Common Exam Questions/Tips

- **Question 1:** What is the atomic number of an atom with 12 protons in its nucleus?

+ Answer: 12

- **Question 2:** What is the difference between atomic number and mass number?

+ Answer: Atomic number is the number of protons, while mass number is the total number of protons and neutrons.

• **Tip:** Make sure to understand the concept of isotopes and how they relate to atomic number and mass number.

## 7. Summary

In summary, the atomic structure is a fundamental concept in chemistry that describes the composition and arrangement of atoms. Key concepts include atomic number, mass number, isotopes, electron configuration, energy levels, and orbitals. Important definitions include proton, neutron, electron, nucleus, and atomic radius. Examples and worked solutions demonstrate how to calculate atomic number and mass number, as well as understand electron configuration. Key points to remember include the importance of atomic number, isotopes, and electron configuration. Common exam questions and tips provide guidance on how to approach exam questions related to atomic structure.

### Revision Checklist:

- Understand the key concepts related to atomic structure
- Know the important definitions related to atomic structure
- Be able to calculate atomic number and mass number
- Understand electron configuration and its importance in chemistry
- Be able to distinguish between isotopes and their relationship to atomic number and mass number

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